

A Simple PromiseBQP-complete Matrix Problem

Pawel M. Wocjan

School of Electrical Engineering and Computer Science

University of Central Florida

Orlando

wocjan@eecs.ucf.edu

Joint work with

Dominik Janzing

Challenges for Quantum Computing

- find new efficient quantum algorithms
(provide more evidence that BQP is strictly greater than BPP)
- understand the computational power and limitations of quantum computers in comparison to classical computers

Efficient Simulation of Quantum Models

- explore what quantum models can be simulated efficiently on a classical computers; see for example:
 - *Match gates and classical simulation of quantum circuits*, Jozsa and Miyake
 - Knill-Gottesman Theorem; *Generalised Clifford groups and simulation of associated quantum circuits*, Clark, Jozsa, Linden
 - Simulating quantum circuits using the formalism of matrix product states and the formalism of contracting tensor networks; generalizations in *On the simulation of quantum circuit*, Jozsa
 - ...

Characterizing BQP

- approximately evaluating the Jones polynomial of plat closures of braids at roots of unity, Aharonov, Jones, Landau
- approximating the Potts model at some points of the Tutte plane, Aharonov, Arad, Eban, Landau
- approximating partition functions of lattice models in certain complex-parameter regimes, Van den Nest, Dür, Raussendorf, Briegel
- approximately evaluating quadratically signed weight enumerators, Knill, Laflamme
- ...

Language / Decision Problem

- a language L is a subset of $\{0, 1\}^*$
- the corresponding decision problem (“language recognition”) consists in determining if
 - either $x \in L$
 - or $x \notin L$

for given input $x \in \{0, 1\}^*$

Complexity Class BQP

- BQP is the class of decision problems that can be solved by a *uniform* family of quantum circuits $\{U_n\}$ in the following sense:

let $x \in \{0, 1\}^n \Rightarrow$ the application of U_n to $|x, 0\rangle$ yields

$$U_n|x, 0\rangle = \alpha_{x,0}|0\rangle \otimes |\psi_{x,0}\rangle + \alpha_{x,1}|1\rangle \otimes |\psi_{x,1}\rangle$$

such that

- $|\alpha_{x,1}|^2 \geq 2/3$ if $x \in L$
- $|\alpha_{x,1}|^2 \leq 1/3$ if $x \notin L$

Promise Problem

- a promise problem Π is a pair $(\Pi_{\text{YES}}, \Pi_{\text{NO}})$ of non-intersecting sets:

$$\Pi_{\text{YES}}, \Pi_{\text{NO}} \subseteq \{0, 1\}^* \text{ and } \Pi_{\text{YES}} \cap \Pi_{\text{NO}} = \emptyset$$

- the set $\Pi_{\text{YES}} \cup \Pi_{\text{NO}}$ is called the *promise*

- standard “language recognition” problems can be cast as the special case:

$$\Pi_{\text{YES}} \cup \Pi_{\text{NO}} = \{0, 1\}^* \text{ (trivial promise)}$$

Complexity Class PromiseBQP

- PromiseBQP is the class of promise problems $(\Pi_{\text{YES}}, \Pi_{\text{NO}})$ that can be solved by a *uniform* family of quantum circuits $\{U_n\}$ in the following sense:

- let $x \in \{0, 1\}^n \Rightarrow$ the application of U_n to $|x, 0\rangle$ yields

$$U_n|x, 0\rangle = \alpha_{x,0}|0\rangle \otimes |\psi_{x,0}\rangle + \alpha_{x,1}|1\rangle \otimes |\psi_{x,1}\rangle$$

such that

- $|\alpha_{x,1}|^2 \geq 2/3$ if $x \in \Pi_{\text{YES}}$
- $|\alpha_{x,1}|^2 \leq 1/3$ if $x \in \Pi_{\text{NO}}$
- nothing is required with respect to $x \notin \Pi_{\text{YES}} \cup \Pi_{\text{NO}}$

Complete Problems

- no complete problems are known for complexity classes such as BPP, BQP, and MA
- complete problems are only known for their promise versions

Diagonal Entry Estimation I

• decide if

• $(A^m)_{jj} \geq g + \epsilon b^m$

• $(A^m)_{jj} \leq g - \epsilon b^m$

• A is a sparse symmetric matrix of size $N \times N$ with $\|A\| \leq b$

• $j \in [1, \dots, N]$

• $m = \text{polylog}(N)$

• $\epsilon = 1/\text{polylog}(N)$

• $g \in [-b^m, b^m]$

Diagonal Entry Estimation II

- the task is to decide if the tuple $(A, b, m, j, \epsilon, g)$ is an element of Π_{YES} or of Π_{NO}
- the instances are defined as follows:

- $(A, b, m, j, \epsilon, g)$ is a YES instance iff

$$(A^m)_{jj} \geq g + \epsilon b^m$$

- $(A, b, m, j, \epsilon, g)$ is a NO instance iff

$$(A^m)_{jj} \leq g - \epsilon b^m$$

- the promise is that only tuples in $\Pi = \Pi_{\text{YES}} \cup \Pi_{\text{NO}}$ are considered

DEE is PromiseBQP-complete

- Diagonal Entry Estimation is PromiseBQP-complete

- we prove this by showing that

- DEE is in PromiseBQP $\Leftrightarrow$

DEE can be solved efficiently on a quantum computer

- DEE is PromiseBQP-hard $\Leftrightarrow$

quantum circuits can be encoded in such a way that we can decide whether they accept an input or not by looking at the corresponding diagonal entry

Spectral Measure I

• let

$$A = \sum_{\lambda} \lambda Q_{\lambda}$$

be the spectral decomposition of A and $|\psi\rangle \in \mathbb{C}^N$ some unit vector

• the spectral measure induced by A and $|\psi\rangle$ is a probability distribution on the spectrum of A such that

$$\Pr(\lambda) = \|Q_{\lambda}|\psi\rangle\|^2$$

• we refer to the process of sampling from the spectral measure as measuring the observable A in the state $|\psi\rangle$

Spectral Measure II

- the expected value of A^m in the state $|\psi\rangle$ is given by

$$\langle\psi|A^m|\psi\rangle = \sum_{\lambda} \lambda^m \langle\psi|Q_{\lambda}|\psi\rangle$$

and is equal to the m th statistical moment of the spectral measure

- we have $(A^m)_{jj} = \langle j|A^m|j\rangle$

$\Rightarrow$ find quantum algorithm for estimating statistical moments of spectral measures to show that DEE is in PromiseBQP

Outline of Quantum Algorithm

- we measure the observable A in the state j
- we obtain the eigenvalue λ and compute λ^m
- we repeat this process and compute the average value of these powers

Measuring the Observable A

- we can measure the observable A by
 - considering A as a Hamiltonian and simulating the corresponding time dynamics

$$U = \exp(iA/\|A\|)$$

- applying the phase estimation algorithm to U

Analysis of Error Sources

- this quantum algorithm works because the following errors can be made sufficiently small with polynomial resources only
 - errors due to imperfections (limited resolution and success probability) of phase estimation
 - statistical errors due to estimation of the expected value from the empirical mean
 - errors due to imperfect simulation of the Hamiltonian time evolution

DEE is PromiseBQP-hard

Equivalent Definition of PromiseBQP

- let U be a BQP quantum circuit and let x be the input
- let p_0 and p_1 denote the probability of finding the output qubit in state $|0\rangle$ and $|1\rangle$, respectively
- elementary calculations show that

$$\langle x, 0 | U^\dagger \sigma_z^{(\text{output})} U | x, 0 \rangle = p_0 - p_1$$

- the sign of this quantity indicates if $x \in \Pi_{\text{YES}}$ or $x \in \Pi_{\text{NO}}$

$$p_0 - p_1 \leq -1/3 \text{ if } x \in \Pi_{\text{YES}} \text{ and } \geq 1/3 \text{ if } x \in \Pi_{\text{NO}}$$

Forward-time Operator

- let $V = U^\dagger \sigma_z^{(\text{output})} U$ and

$$V = V_{M-1} V_{M-2} \cdots V_0$$

the decomposition of V into elementary gates (M is odd)

- the forward-time operator F is

$$F = \sum_{\ell=0}^{M-1} |\ell+1\rangle\langle\ell|_{\text{clock}} \otimes V_\ell$$

- we have $F^{2M} = I$ because $V^2 = I$

Orbit

- we restrict our attention to the orbit

$$\left\{ F^\ell |0\rangle_{\text{clock}} \otimes |x, 0\rangle : \ell \in \mathbb{N} \right\}$$

Action of the Forward-time Operator

- the input x is fully rejected ($p_0 = 1$) $\Rightarrow$
 - the orbit is M -periodic
 - the action of F is isomorphic to the action of a cyclic shift in M dimensions

$$|\ell\rangle \mapsto |(\ell + 1) \bmod M\rangle$$

where $F^\ell |0\rangle \otimes |x, 0\rangle$ corresponds to $|\ell\rangle$

- the spectral measure $R^{(0)}$ induced by F and $|0\rangle \otimes |x, 0\rangle$ is the uniform distribution on the M th roots of unity

$$\Pr \left(\exp(i\pi 2\ell/M) \right) = \frac{1}{M}$$

Action of the Forward-time Operator

- the input x is fully accepted ($p_1 = 1$) $\Rightarrow$
 - the action of F is isomorphic to the action of a cyclic shift in M dimensions with an **additional phase -1**

$$\begin{aligned} |\ell\rangle &\mapsto |\ell + 1\rangle \text{ for } \ell = 0, \dots, M - 2 \\ |M - 1\rangle &\mapsto -|0\rangle \end{aligned}$$

- the spectral measure $R^{(1)}$ induced by F and $|0\rangle \otimes |x, 0\rangle$ is

$$\Pr \left(\exp(i\pi(2\ell + 1)/2M) \right) = \frac{1}{M}$$

Spectral Measure

- in the general case, the spectral measure induced by F and $|0\rangle \otimes |x, 0\rangle$ is a convex combination

$$R = p_0 R^{(0)} + p_1 R^{(1)}$$

Sparse Matrix

- we define the self-adjoint sparse matrix

$$A = \frac{1}{2}(F + F^\dagger)$$

- the spectral measure induced by A and $|0\rangle \otimes |x, 0\rangle$ is given by

$$P = p_0 P^{(0)} + p_1 P^{(1)}$$

where the supports of $P^{(0)}$ and $P^{(1)}$ are given by the real parts of $R^{(0)}$ and $R^{(1)}$ (cosine values)

Spectral Measures

- $p_0 = 1 \Rightarrow$ the spectral measure $P^{(0)}$ has the form

	for $\ell = 1, \dots, M - 1$
$\lambda^{(0)} = +1$	$\lambda_{\ell}^{(0)} = +\cos(2\pi\ell/M)$
$\Pr(\lambda_0^{(0)}) = 1/M$	$\Pr(\lambda_{\ell}^{(0)}) = 2/M$

- $p_1 = 1 \Rightarrow$ the spectral $P^{(1)}$ has the form

	for $\ell = 1, \dots, M - 1$
$\lambda^{(1)} = -1$	$\lambda_{\ell}^{(1)} = -\cos(2\pi\ell/M)$
$\Pr(\lambda_0^{(1)}) = 1/M$	$\Pr(\lambda_{\ell}^{(1)}) = 2/M$

Statistical Moments

- identify j with the label of the initial state $|0\rangle \otimes |x, 0\rangle$
- the m th statistical moment is equal to

$$(A^m)_{jj} = (p_0 - p_1) \cdot \left(1 + \sum_{\ell=1}^{M-1} \cos^m(2\pi\ell/M) \right)$$

- for $m = M^3 = \text{poly}(\# \text{ quantum gates})$, we have

$$(A^m)_{jj} \approx p_0 - p_1$$

Strengthening of the Result

- the DEE problem for sparse matrices A with entries restricted to $0, \pm 1$ remains PromiseBQP-hard
- the problem of estimating off-diagonal terms is also in PromiseBQP
- if A is restricted to have only $0, 1$ as entries, then the problem of estimating differences

$$(A^m)_{ii} - (A^m)_{ij}$$

is PromiseBQP-complete

- the adjacency matrix A can be obtained from simple string rewriting rules (more explicit than saying that A is sparse)

Connection to Quantum Algorithms

quantum algorithms that

- implement

$$\exp(-iAt)$$

- and compute

$$\langle \psi | f(A) | \psi \rangle$$

for sparse A and smooth functions f

- *Quantum algorithm for solving linear systems of equations, Harrow, Hassidim, Lloyd*
- *A quantum algorithm to solve nonlinear differential equations, Leyton, Osborne*

Conclusions

- a purely classical (quantum free) characterization of BQP in terms of a simple matrix problem
- quantum computers are better at counting and comparing the number of solutions of combinatorial problems (provided that $BQP \neq P$)
- $\Rightarrow$ can we find new quantum algorithms based on this observation?